

RESEARCH ARTICLE



# JPEG compression-resistant adversarial attack with invisible watermark embedding

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## ABSTRACT

Deep neural networks have achieved impressive performance across various domains but remain highly vulnerable to adversarial attacks. Existing adversarial methods often introduce imperceptible perturbations as meaningless noise, which become ineffective after standard image compression. In this paper, we propose a novel JPEG compression-resistant adversarial attack framework that simultaneously embeds invisible watermark information and adversarial perturbations directly into the DCT domain. By introducing a simulated JPEG recompression preprocessing layer and employing a straight-through estimator, our method enables gradient backpropagation through quantized DCT coefficients. Targeted modifications are performed based on saliency maps, while the number of coefficient alterations is strictly constrained to minimize image distortion. Experimental results demonstrate that the proposed approach achieves robust adversarial attacks and accurate watermark recovery even after standard compression, outperforming traditional adversarial methods.

## ARTICLE HISTORY

Received 17 December 2025  
Accepted 9 March 2026

## KEYWORDS

Adversarial attack; JPEG compression; watermarking; deep neural network; saliency map

## Introduction

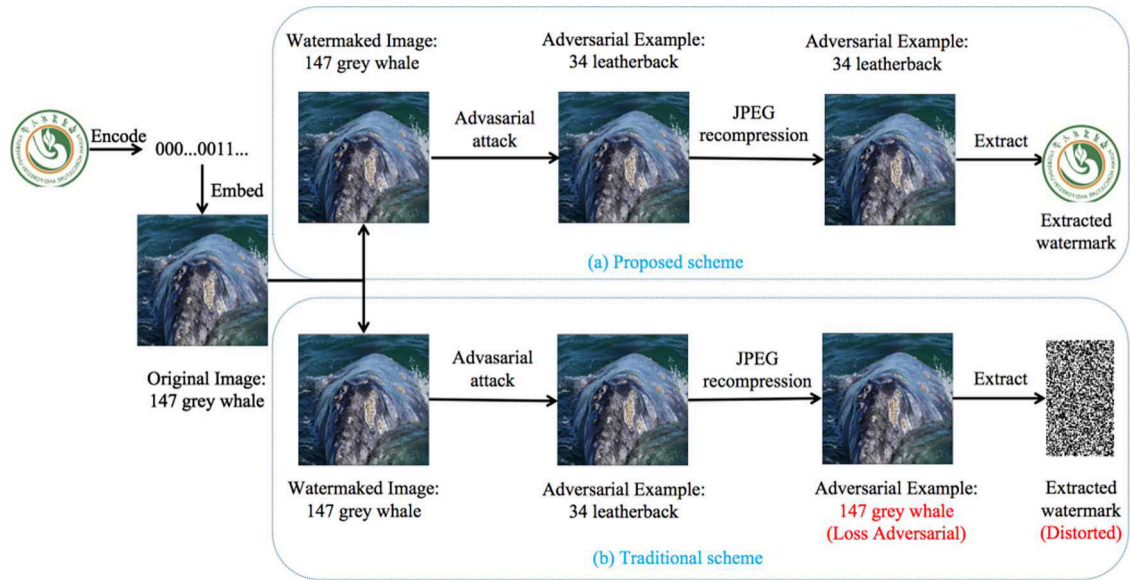
Deep neural network (DNN) have achieved remarkable accuracy on many complex datasets, enabling neural networks to perform at or even surpass human-level capabilities across a variety of tasks [1]. However, despite their high accuracy on test sets, DNN models remain notably vulnerable to adversarial attacks [2]. Numerous studies [3–5] have explored adversarial attack methods that aim to generate adversarial examples by introducing subtle perturbations to original images, causing the predicted labels to differ from the true labels. More recently, advanced adversarial attack techniques [6,7] have focused on reducing the perceptibility of perturbations while improving attack success rates. These methods generate perturbations that are virtually imperceptible to the human eye yet can effectively cause misclassification. Nevertheless, existing adversarial attack methods often overlook the intrinsic characteristics of images, and the perturbations introduced tend to be random, meaningless noise.

To overcome these limitations, recent research [8–10] has shifted towards embedding meaningful perturbations into original images, thereby combining adversarial attacks with invisible watermarking in a unified task. Invisible watermarking, known for its imperceptibility and forensic traceability [11], has emerged as a robust solution for copyright protection. However, a practical challenge remains:

adversarial examples are typically generated as floating-point tensor encodings, whereas standard raster images are stored as integer matrices. For real-world applications, it is necessary to generate adversarial examples with integer-valued perturbations within bounded ranges. This issue becomes particularly critical when dealing with JPEG images [12]. JPEG compression quantizes DCT coefficients of the YCbCr colour channels and discards high-frequency components to reduce file size.

While this discarded information may minimally affect model classification on clean images, it can severely degrade adversarial perturbations, as shown in Figure 1(b). Adversarial examples often lose their effectiveness after JPEG compression, particularly when the compression quality is low. Some studies [13–15] have even employed JPEG compression as a defense mechanism against adversarial attacks. These limitations reveal a clear research gap: how to design adversarial attacks that remain effective after JPEG compression while preserving visual quality and reliable watermark extraction. Addressing this gap is crucial for real-world applications such as social media image sharing, copyright protection, and defense against unauthorized data scraping, where recompression is unavoidable.

To address these challenges, this paper proposes a novel JPEG compression-resistant adversarial



**Figure 1.** Examples of the adversarial examples generated by the proposed and traditional method.

attack with invisible watermark embedding, where both adversarial perturbations and watermark information are directly embedded in the DCT domain. Specifically, a simulated JPEG recompression preprocessing layer is introduced before feeding the image into the target model. Using a straight-through estimator, gradients are propagated through quantized DCT coefficients, enabling direct and targeted modifications based on saliency maps. The adversarial example generated by the proposed method successfully retains both adversarial properties and watermark extraction capability even after JPEG compression, as shown in Figure 1(a). The main contributions of this paper are summarized as follows:

- (1) We propose a novel JPEG compression-resistant adversarial attack framework with invisible watermark embedding, in which adversarial perturbations and watermark information are jointly embedded directly into the DCT domain, ensuring inherent compatibility with JPEG compression.
- (2) A differentiable JPEG recompression preprocessing layer is introduced, together with a straight-through estimator, enabling effective gradient backpropagation through quantized DCT coefficients and allowing adversarial optimization under realistic compression constraints.
- (3) A saliency-guided coefficient modification strategy is designed, which strictly limits the number and magnitude of modified DCT coefficients, achieving a favourable trade-off between attack success, visual quality, and watermark robustness.
- (4) Extensive experiments demonstrate that the proposed method consistently outperforms existing approaches in terms of post-compression attack success rate, image quality, and

watermark extraction accuracy, validating its practical effectiveness in real-world compressed image scenarios.

## Related work

Adversarial attacks are crafted perturbations added to input data with the intent of misleading machine learning models. Common adversarial attack methods include FGSM [3], DeepFool [4], C&W [16], and JSMA [17]. Building on these foundational techniques, several improved adversarial attack methods have been proposed. BIM [18] extends FGSM with multiple iterations and gradient clipping to generate more controlled perturbations. PGD [5] further enhances BIM by projecting perturbations into a defined  $L_p$ -norm ball, making it one of the strongest first-order adversarial attacks. MI-FGSM [19] introduces a momentum term to stabilize the direction of gradient updates and escape poor local minima. In addition to gradient-based attacks, generative adversarial network [20] based approaches have also been developed for crafting adversarial examples. These methods leverage generative models to produce diverse and transferable adversarial perturbations.

The continued evolution of adversarial attack algorithms not only deepens our understanding of deep neural network vulnerabilities but also poses ongoing challenges to adversarial defense strategies. Studying adversarial examples contributes to both model robustness and intellectual property protection by helping prevent unauthorized model usage. However, existing adversarial attack research often overlooks the practical issue of storage format, such as JPEG. JPEG compression is one of the most widely adopted image coding standards in practical image

storage and transmission. It operates by first converting images into the YCbCr colour space and then applying block-wise discrete cosine transform (DCT) on non-overlapping  $8 \times 8$  blocks. The resulting DCT coefficients represent different frequency components, where low-frequency coefficients preserve the main image structure and high-frequency coefficients capture fine details. To achieve compression, JPEG applies quantization to these coefficients, aggressively suppressing high-frequency components that are less perceptually important. While this process has limited impact on human visual perception, it can significantly distort or eliminate subtle adversarial perturbations optimized in the spatial domain.

When adversarial examples are compressed by JPEG, their adversarial properties can degrade significantly, rendering them ineffective in real-world deployment scenarios such as social media platforms. To address this challenge, recent efforts have focused on compression-resistant adversarial attacks, aiming to preserve adversarial effectiveness after typical image compression encountered in real applications. Several methods have been proposed to enhance robustness under such conditions. Hu et al. propose ST-IFGSM [21] with  $L_0$  regularization to detect the vulnerability and enhance the robustness of adversarial examples. To support both adversarial attack on deep learning model and copyright protection, representative approaches that embed invisible watermarks into adversarial examples are also investigated. Liang et al. propose BHI [9], an adversarial example generation method that embeds invisible watermarks based on a Basin-Hopping optimization strategy, effectively balancing imperceptibility and attack success. It was further improved by ISWP, a high-fidelity adversarial attack technique that integrates secure and invisible watermark perturbations to enhance stealthiness and preserve semantic consistency of the adversarial examples [10]. Lin et al. propose GMM, embeds information into adversarial perturbations by selectively blocking the updating noise at specific positions using a message mask encoded from the information. This method enables processed images to exhibit adversarial properties while simultaneously serving as carriers of information [22]. Similarly, robust image watermarking schemes using fine-grained differentiable JPEG compression modules have been explored to improve watermark preservation under multiple real-world distortions, demonstrating the practical value of compression-aware optimization [23]. There are also related frequency-domain defense strategies that employ DCT coefficient manipulation to protect against malicious image editing while maintaining robustness to JPEG quantization [24]. Beyond JPEG compression, recent analyses discuss vulnerabilities and robustness trade-offs in watermark design, such as leakage in robust watermarks via channel-aware

feature extraction [25]. Semi-blind deep learning-based JPEG-resistant watermarking techniques have been proposed to achieve high imperceptibility and robustness across quality factors using generative networks [26]. In addition, several recent studies have explored related topics in adversarial robustness, multimedia security, and transform-domain image protection [27–30].

However, the previous methods still exhibit limited robustness against aggressive JPEG compression, which is commonly applied when images are uploaded to social platforms. From an adversarial attack perspective, this compression pipeline exposes a key limitation of spatial-domain and transform-agnostic perturbations: they are not optimized with respect to the quantization and rounding operations inherent in JPEG encoding. Other transform domains, such as wavelet-based or feature-level representations, do not directly correspond to the JPEG compression process and therefore cannot accurately model the effect of recompression. In contrast, operating directly in the DCT domain enables explicit control over frequency components that survive JPEG quantization, making it possible to design adversarial perturbations that are inherently robust to recompression. This alignment between attack generation and the JPEG encoding mechanism is the primary motivation for adopting DCT-domain processing in this work.

## Proposed method

In this section, we propose a JPEG compression-resistant adversarial attack method with invisible watermark embedding, which not only maintains high attack success rates after recompression but also ensures reliable watermark extraction. The overall framework proposed in this paper is shown in Figure 2.

### Watermark embedding

During the watermark embedding phase, the watermark is first encoded into a bitstream and embedded into the cover image. Specifically, the dequantized DCT coefficients  $X$  of the cover image are computed. The 6, 7, and 8 diagonals of each DCT block is selected as the embedding domain  $E_{678}$ , as illustrated in Figure 2. In an  $8 \times 8$  DCT block, different diagonals correspond to different frequency bands with distinct perceptual and compression characteristics. In this work, the 6th–8th diagonals are selected for watermark embedding because they lie in the mid-to-high frequency range that is relatively robust to JPEG quantization while having limited impact on visual quality, enabling reliable watermark extraction after recompression. Conversely, the 3rd–5th diagonals are chosen as the adversarial attack domain, as coefficients in this range exert stronger influence on deep neural



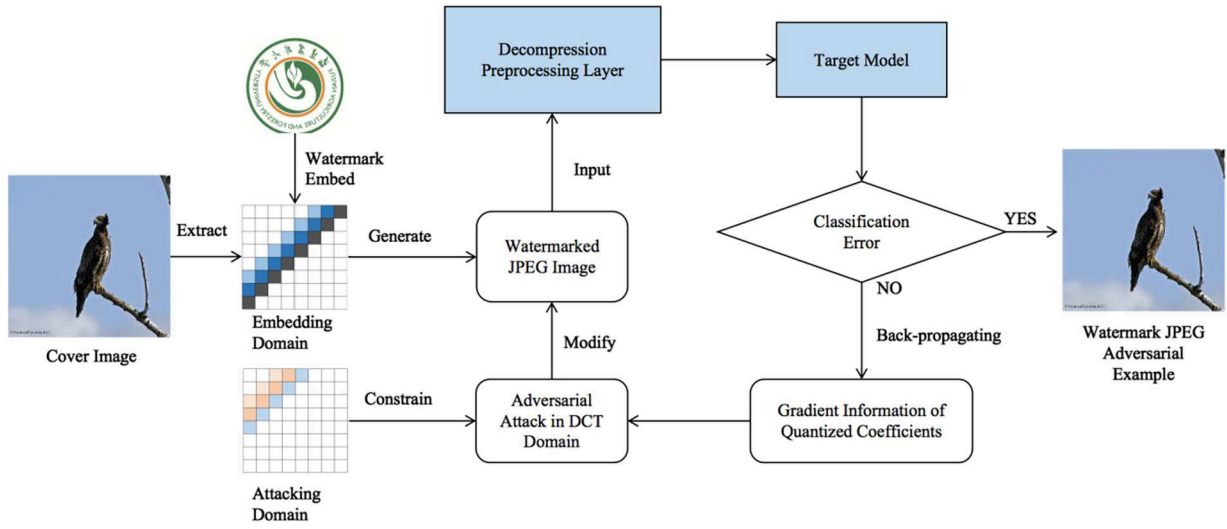


Figure 2. Overall framework of the proposed method.

network predictions while remaining less visually salient than low-frequency components. Then, the J-UNIWARD distortion cost  $p_{ij}$  is calculated for each element of  $X$  based on Equation (1).

$$\rho_{ij} = \sum_{k=1}^3 \sum_{u=1}^{n_1} \sum_{v=1}^{n_2} \frac{|W_{uv}^{(k)}(J^{-1}(X)) - W_{uv}^{(k)}(J^{-1}(X_{s_{ij}}))|}{|W_{uv}^{(k)}(J^{-1}(X))| + \sigma} \quad (1)$$

where  $i, j$  denotes the row and column indices of the DCT matrix.  $W_{uv}^{(k)}$  denotes the wavelet coefficient at position  $(u, v)$  in the  $k$ -th subband of the first-level decomposition, and  $X_{s_{ij}}$  represents the modified version of  $X$ . The small constant  $\sigma = 2^{-6}$  is introduced to avoid division by zero. Furthermore,  $J^{-1}(X)$  denotes the spatial image reconstructed from  $X$ , and  $n_1, n_2$  are the dimensions of the cover image.

Apply mean filtering and DCT transformation to the original spatial image to obtain  $\bar{X}$ . Then, compute the asymmetric distortion measures  $\rho^+$  and  $\rho^-$  using Equation (2):

$$\rho^+ = \begin{cases} \alpha \cdot \rho_{ij}, & x_{ij} < \frac{\bar{x}_{ij}}{q_{ij}} \\ \rho_{ij}, & x_{ij} \geq \frac{\bar{x}_{ij}}{q_{ij}} \end{cases} \quad (2)$$

$$\rho^- = \begin{cases} \alpha \cdot \rho_{ij}, & x_{ij} > \frac{\bar{x}_{ij}}{q_{ij}} \\ \rho_{ij}, & x_{ij} \leq \frac{\bar{x}_{ij}}{q_{ij}} \end{cases}$$

where  $x_{ij}$  and  $\bar{x}_{ij}$  are elements from  $X$  and  $\bar{X}$ , respectively,  $q_{ij}$  is the quantization step and  $\alpha$  is a parameter typically set to 0.7.

Next, the message  $m$  is encoded using Reed Solomon (RS) codes and shuffled in blocks of  $\log_2(n+1)$  bits to produce the encoded message  $m'$ . The encoded message  $m'$  is then embedded into the cover sequence of  $X'$ , using ternary STC (Syndrome-

Trellis Codes) coding with  $\rho^+$  and  $\rho^-$  to generate the stego sequence  $S$  of the stego image.

### Image decompression preprocessing

To generate JPEG adversarial examples directly, our attack method modifies the quantized DCT coefficients of the JPEG image. This requires implementing a pre-processing layer that simulates decompression to achieve  $f_{Q \rightarrow P}: X_Q \rightarrow X_P$ ,

where  $X_Q \in [[-1024, 1023]]^{N \times M \times 3}$  represents the quantized DCT coefficients of the YCbCr channels of the original image with dimensions of  $N \times M$ , and  $X_P \in [[0, 255]]^{N \times M \times 3}$  denotes the RGB pixel image decompressed into the spatial domain. This process is accomplished by applying Equations (3–6) with the colour channel transformation to each  $8 \times 8$  block of  $X_Q$ .

$$D = D_Q \times Q \quad (3)$$

$$P'_D = \text{ROUND}(\text{IDCT}(D)) + 128 \quad (4)$$

$$P_D = \text{CUT}(P'_D) \quad (5)$$

$$\text{ROUND} \left( \begin{pmatrix} 1 & 0 & 1.402 \\ 1 & -0.34414 & -0.71414 \\ 1 & 1.772 & 0 \end{pmatrix} \times \begin{pmatrix} X_Y \\ X_{Cb} - 128 \\ X_{Cr} - 128 \end{pmatrix} \right) \approx \begin{pmatrix} X_R \\ X_G \\ X_B \end{pmatrix} = X_P \quad (6)$$

where the quantized DCT coefficient matrix of a block in a grayscale image, quantized with a given quantization table  $Q$ , is denoted as  $D_Q$ .  $P_D$  is the spatial pixel matrix,  $\text{IDCT}()$  is the inverse process of DCT,  $\text{ROUND}()$  is the rounding operation, and  $\text{CUT}()$  is used to truncate values exceeding 255 or falling below 0.



### Adversarial attack in DCT domain

Adversarial attack in DCT domain is the main procedure of the proposed scheme, which modifies the coefficients in the attack domain of the watermarked image until the target model misclassifies the image. Firstly, the input image  $X$  of size  $N \times M \times 3$  undergoes classification through the model  $F()$ , as illustrated in Equation (7).

$$\text{label}(X) = \underset{j}{\operatorname{argmax}} F_j(X) \quad (7)$$

where  $F_j(X)$  is the model output value for the  $j$ -th label component.

At this point, the partial derivative of  $F_j(X)$  with respect to the element  $x_p$  at position  $p$  in the image input is  $\frac{\partial F_j(X)}{\partial x_p}$ . Assuming we want the model's classification result for the image to be  $t$ , the allowable attack domain is represented as  $E_{adv} \in \llbracket 0, 1 \rrbracket^{N \times M \times 3}$ . We first construct the saliency maps in two directions,  $S^+(X, t)$  and  $S^-(X, t)$ , as shown in Equations (8–9). When constructing the saliency maps, we only calculate the partial derivatives for labels with confidence levels higher than the target label, which significantly reduces the number of backpropagation calculations.

#### Algorithm DJSMA: JSMA Attack in the DCT Domain

**Input:** Target model  $F$  with preprocessing layers, quantized DCT coefficients of the image  $X$ , attack region  $E_{adv}$ , number of iterations  $\tau$ ,  $l_\infty$ -norm constraint  $\mu$ , Target label  $t$ .

```

1: Procedure DJSMA ( $F, X, E_{adv}, \tau, \mu, t$ )
2:    $mask = E_{adv}$   $\triangleright$ Record the position of the adversarial perturbation
3:    $change = X \times 0$   $\triangleright$ Track the modification count for each position
4:    $X_0 = X$ 
5:   For  $i = 0$  to  $\tau$ :  $\triangleright$ Iteratively perform adversarial attacks
6:     If  $\arg \max F_j(X_i) = t$  then  $\triangleright$ Targeted attack successful
7:        $X \leftarrow X_i$   $\triangleright$ Store the adversarial example, terminate the attack
8:       Break
9:     Else
10:      Compute the saliency maps  $S^+(X_i, t)$  and  $S^-(X_i, t)$ 
11:       $S^+ = S^+ \times mask$ 
12:       $S^- = S^- \times mask$ 
13:       $p^+ = \arg \max S^+[k]$   $\triangleright$ Record the position with the maximum positive perturbation
14:       $p^- = \arg \max S^-[k]$   $\triangleright$ Record the position with the maximum negative perturbation
15:      If  $S^+[p^+] > S^-[p^-]$  then  $\triangleright$ Select the optimal perturbation position
16:         $p = p^+, sign = 1$ 
17:      Else
18:         $p = p^-, sign = -1$ 
19:      End If
20:       $X_0[p] = X_0[p] + sign$   $\triangleright$ Apply the perturbation
21:       $change[p] = change[p] + 1$   $\triangleright$ Update the modification count for that position
22:      If  $change[p] > \mu$  then  $\triangleright$ Exclude from further updates if the allowed modification count is exceeded
23:         $mask[p] = 0$ 
24:      End If
25:      End If
26:       $X_{i+1} = X_i$ 
27:    End For
28:    return  $X$ 

```

(Continued)

Continued.

#### Algorithm DJSMA: JSMA Attack in the DCT Domain

$$S^+(X, t)[p] = \begin{cases} 0, \frac{\partial F_t(X)}{\partial x_p} < 0 \text{ or } \sum_{j: F_j(X) > F_t(X)} \frac{\partial F_j(X)}{\partial x_p} > 0 \text{ or } p \in E_{adv} \\ \frac{\partial F_t(X)}{\partial x_p} \times \left| \sum_{j: F_j(X) > F_t(X)} \frac{\partial F_j(X)}{\partial x_p} \right|, \text{ other} \end{cases} \quad (8)$$

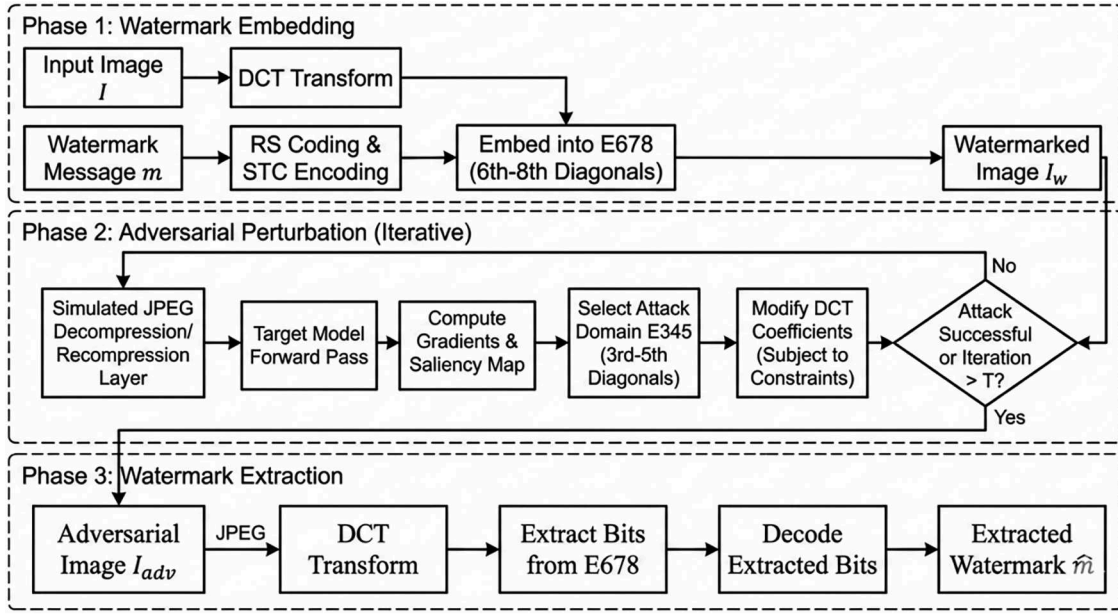
$$S^-(X, t)[p] = \begin{cases} 0, \frac{\partial F_t(X)}{\partial x_p} \geq 0 \text{ or } \sum_{j: F_j(X) > F_t(X)} \frac{\partial F_j(X)}{\partial x_p} \leq 0 \text{ or } p \in E_{adv} \\ \left| \frac{\partial F_t(X)}{\partial x_p} \right| \times \sum_{j: F_j(X) > F_t(X)} \frac{\partial F_j(X)}{\partial x_p}, \text{ other} \end{cases} \quad (9)$$

The detailed adversarial attack in the DCT domain of the proposed scheme is shown in Algorithm 1. Based on the calculation of the saliency map, the proposed method is conducted by modifying the value at only one position at a time. The attack domain  $E_{adv}$  restricts the allowed modifications during the adversarial attack, while the number of iterations  $\tau$  limits the  $l_0$  norm loss of the adversarial examples, and  $\mu$  restricts the  $l_\infty$  norm loss. Additionally, since the algorithm allows modifications in both  $\pm 1$  directions, there is a possibility of the algorithm entering an infinite loop at the same position. To prevent this situation, the variable *change* is used to count the number of modifications at each position.

To provide a clearer roadmap of how the watermark and adversarial perturbations are decoupled in the DCT domain, Figure 3 presents a holistic flowchart of the entire framework. It systematically integrates the watermark embedding stage, the differentiable JPEG-simulated adversarial attack process, and the subsequent watermark extraction phase. Specifically, the figure illustrates the complete pipeline: starting from the initial DCT-based watermark embedding in high-frequency diagonal coefficients, proceeding to the iterative saliency-guided adversarial perturbation applied to mid-frequency diagonals, and finally performing watermark extraction after JPEG compression. This integrated visualization clarifies the functional separation of embedding and attack domains while highlighting their coordinated operation within the unified framework.

### Experimental results

All experiments were conducted using ImageNet dataset with six pre-trained neural networks: Inception\_V3, SqueezeNet, ResNet101, VGG19, ShuffleNet\_V2, and ResNet152. To simulate practical conditions on social media platforms, the adversarial examples were subjected to a second round of JPEG compression with quality factor  $Q = 75$ . The proposed method operates within a constrained attack domain, with a specified iteration limit ( $\tau = 1500$ ) and a



**Figure 3.** The complete pipeline of the proposed JPEG compression-resistant adversarial attack and watermark embedding framework.

maximum allowed perturbation magnitude ( $\mu = 1$ ) for each quantization coefficient. We selected three representative methods for comparison: Adv-watermark [8], ST-IFGSM [21], BHI [9], ISWP [10] and GMM [22]. All parameter settings followed the configurations specified in the original papers.

### Attack performance comparison

In this section, adversarial examples are crafted based on watermarked images, followed by a single JPEG compression to simulate practical transmission scenarios. Table 1 presents the classification accuracy (%) across six commonly used CNN architectures for original images, watermarked images, and adversarial examples generated by various methods.

To evaluate attack performance, we compute the attack success rate as the difference between classification accuracy on watermarked images and on adversarial examples. The results in Table 1 demonstrate that the proposed method consistently achieves the highest attack success rate across all models, confirming its superior robustness against JPEG compression. For instance, on VGG19, the success rate reaches 62.2% (63.4% to 1.2%), outperforming all

baselines. Adv-watermark and ST-IFGSM shows improved attack performance compared to original watermarked image due to its finer perturbation quantization, which helps retain adversarial effectiveness after compression. BHI shows moderate success, achieving high attack rates but still lower than others methods. Its perturbation is more compression-resistant but less optimized for minimal visibility. ISWP performs better than BHI and ST-IFGSM on all models except InceptionV3, likely due to its adaptive attack in both spatial and frequency domains.

The superior attack performance of the proposed method can be attributed to its explicit alignment with the JPEG compression pipeline. Unlike spatial-domain attacks, whose perturbations are largely destroyed by quantization and rounding, the proposed method directly modifies quantized DCT coefficients that survive JPEG recompression. In addition, the saliency-guided coefficient selection ensures that perturbations are concentrated on frequency components that have a stronger influence on model decision boundaries. This combination enables the proposed method to maintain high attack success rates even after recompression, whereas baseline methods suffer from significant performance degradation.

**Table 1.** Model classification accuracy (%) for different image sources after JPEG compression.

|                    | Vgg19      | Inceptionv3 | ResNet101  | ResNet152  | SqueezeNet | ShuffleNet |
|--------------------|------------|-------------|------------|------------|------------|------------|
| Original images    | 64.8       | 66          | 71.9       | 73.3       | 52.1       | 63.3       |
| Watermarked images | 63.4       | 65.8        | 71.7       | 72.2       | 48.5       | 59.9       |
| Adv-watermark      | 3.2        | 5.8         | 5.3        | 6          | 3.1        | 4.3        |
| ST-IFGSM           | 3.4        | 9.3         | 6.4        | 7.2        | 3.8        | 6.3        |
| BHI                | 15.5       | 20.1        | 18.3       | 18.2       | 15.2       | 20.2       |
| ISWP               | 2.1        | <b>4.3</b>  | <b>3.6</b> | 4.8        | 1.5        | 2.5        |
| GMM                | 3.5        | 6.2         | 4.3        | 4.5        | 2.4        | 2.8        |
| Proposed           | <b>1.2</b> | 4.6         | 3.9        | <b>4.6</b> | <b>0.6</b> | <b>0.8</b> |

**Table 2.** Comparison of average PSNR values (dB).

|                    | Vgg19        | Inceptionv3  | ResNet101    | ResNet152    | SqueezeNet   | ShuffleNet   |
|--------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Watermarked images | 37.81        | 37.81        | 37.81        | 37.81        | 37.81        | 37.81        |
| Adv-watermark      | 26.15        | 25.96        | 26.42        | 26.16        | 26.83        | 25.43        |
| ST-IFGSM           | 36.57        | 36.00        | 36.31        | 36.29        | 37.19        | 36.96        |
| BHI                | 33.47        | 33.93        | 33.21        | 33.45        | 34.1         | 33.85        |
| ISWP               | 30.36        | 30.54        | 31.01        | 30.45        | 30.83        | 30.21        |
| GMM                | <b>48.42</b> | <b>48.45</b> | <b>48.42</b> | <b>48.48</b> | <b>48.51</b> | <b>48.42</b> |
| Proposed           | 36.56        | 36.91        | 36.01        | 36.02        | 37.11        | 36.58        |

### Visual quality comparison

In addition to attack effectiveness, the visual quality of adversarial examples is a critical consideration, especially in scenarios involving watermarking and perceptual integrity. To evaluate this, two standard image quality metrics are used: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). The average PSNR and SSIM values for adversarial images generated under different attack methods are presented in Tables 2 and 3, respectively. The watermarked images serve as the reference baseline.

The results indicate that the proposed method maintains high visual fidelity, with PSNR values consistently above 36 dB and SSIM values above 0.94 across all models. This demonstrates that the perturbations introduced are both subtle and structurally consistent with the original watermarked images. The high PSNR and SSIM values achieved by the proposed method are primarily due to the constrained modification strategy in the DCT domain. By limiting perturbations to carefully selected mid-frequency coefficients and strictly controlling the number of modified positions, the proposed method avoids disturbing low-frequency components that dominate visual perception. Compared to GMM, the proposed method achieves lower PSNR and SSIM values, while still delivering significantly better attack success rates. This highlights the advantage of the proposed method in balancing robustness and imperceptibility. BHI exhibits moderate degradation in visual quality. Although less disruptive than ISWP, its structural impact is more noticeable than that of the proposed and ST-IFGSM methods. Adv-watermark shows the lowest visual quality among all methods. In contrast, Adv-watermark introduces substantially stronger perturbations in the spatial domain, leading to significantly lower PSNR and SSIM values, which may reduce perceptual quality under human inspection. While ISWP achieves reasonable

attack success, it comes at the cost of significantly reduced perceptual quality, which may expose the adversarial nature of the image under human inspection or watermark detection algorithms.

### Watermark extraction performance comparison

In adversarial watermarking applications, maintaining the integrity of the embedded watermark is essential. An effective attack method must not only mislead the classifier but also ensure that the adversarial perturbations do not destroy the embedded watermark, especially after JPEG compression. Table 4 summarizes the average watermark extraction error rates (%) across six different CNN architectures after JPEG compression and adversarial attacks.

The results show that the proposed method achieves the lowest watermark extraction error rates across all models, indicating excellent compatibility with the embedded watermark and strong robustness against JPEG compression. This suggests that the adversarial perturbations generated by the proposed method are well-aligned with the watermark-preserving regions of the image. Compared with ST-IFGSM, the proposed method reduces the extraction error rate by approximately 50% on average. This demonstrates that the proposed method introduces less distortion to watermark-relevant regions while maintaining high attack success. BHI and GMM show the worst performance in terms of watermark preservation. This indicates that the perturbation patterns generated by them significantly interfere with the embedded watermark, making it unsuitable for applications requiring both classification evasion and watermark integrity. Adv-watermark also exhibits relatively high extraction error rates after JPEG compression, indicating that spatial-domain perturbations without explicit frequency-domain decoupling may interfere

**Table 3.** Comparison of average SSIM values.

|                    | Vgg19        | Inceptionv3  | ResNet101    | ResNet152    | SqueezeNet   | ShuffleNet   |
|--------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Watermarked images | 0.981        | 0.981        | 0.981        | 0.981        | 0.981        | 0.981        |
| Adv-watermark      | 0.831        | 0.818        | 0.824        | 0.852        | 0.816        | 0.84         |
| ST-IFGSM           | 0.943        | 0.960        | 0.932        | 0.934        | 0.962        | 0.951        |
| BHI                | 0.969        | 0.954        | 0.948        | 0.931        | 0.954        | 0.961        |
| ISWP               | 0.878        | 0.901        | 0.892        | 0.924        | 0.913        | 0.893        |
| GMM                | <b>0.974</b> | <b>0.975</b> | <b>0.979</b> | <b>0.977</b> | <b>0.974</b> | <b>0.977</b> |
| Proposed           | 0.958        | 0.965        | 0.942        | 0.941        | 0.966        | 0.952        |



**Table 4.** Average watermark extraction error rates (%) after JPEG compression.

|               | Vgg19       | Inceptionv3 | ResNet101   | ResNet152   | SqueezeNet  | ShuffleNet  |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Adv-watermark | 13.26       | 14.32       | 13.43       | 13.76       | 12.78       | 11.01       |
| ST-IFGSM      | 9.37        | 10.28       | 12.28       | 13.46       | 9.43        | 5.44        |
| BHI           | 32.24       | 35.23       | 34.43       | 35.23       | 33.65       | 34.21       |
| ISWP          | 12.87       | 13.74       | 13.45       | 13.26       | 12.76       | 13.22       |
| GMM           | 35.43       | 36.33       | 37.21       | 37.54       | 35.66       | 36.27       |
| Proposed      | <b>3.79</b> | <b>4.16</b> | <b>5.27</b> | <b>5.71</b> | <b>3.75</b> | <b>2.18</b> |

**Table 5.** Performance of the proposed method under representative JPEG quality factors (QF = 75 and 85).

| Model                      | Vgg19 | Inceptionv3 | ResNet101 | Vgg19 | Inceptionv3 | ResNet101 |
|----------------------------|-------|-------------|-----------|-------|-------------|-----------|
| QF                         | 75    | 75          | 75        | 85    | 85          | 85        |
| Classification Accuracy(%) | 1.2   | 4.6         | 3.9       | 3.9   | 6.5         | 5.5       |
| PSNR                       | 36.56 | 36.91       | 36.01     | 40.16 | 40.42       | 39.53     |
| SSIM                       | 0.958 | 0.965       | 0.942     | 0.98  | 0.982       | 0.972     |
| Extraction Error rate(%)   | 3.79  | 4.16        | 5.27      | 4.57  | 5.62        | 5.81      |
| Runtime(s)                 | 1.05  | 0.88        | 0.85      | 1.01  | 0.86        | 0.82      |

with watermark integrity. ISWP exhibits moderate performance with error rates, however, it is designed to be adaptive in spatial regions. Its perturbation strength appears to conflict with watermark embedding, leading to substantial watermark degradation.

The robustness of watermark extraction in the proposed method benefits from the frequency-domain decoupling between watermark embedding and adversarial perturbation. Since watermark information is embedded in relatively stable DCT coefficients and adversarial perturbations are applied to a separate frequency range, mutual interference is effectively minimized. Baseline methods do not explicitly consider this decoupling, leading to higher watermark extraction error rates after JPEG compression. This design choice allows the proposed method to simultaneously preserve watermark integrity and adversarial effectiveness.

### Robustness and efficiency under different JPEG quality factors

To further examine the robustness of the proposed method under different JPEG compression settings, we additionally report experimental results under two representative JPEG quality factors, QF = 75 and QF = 85. These two quality factors correspond to moderate and mild compression levels that are commonly encountered in practical image storage and sharing scenarios.

As shown in Table 5, the proposed method maintains stable attack success rates and high visual quality under both compression settings. The consistent performance across QF = 75 and QF = 85 indicates that the proposed JPEG-aware DCT-domain design is robust to typical JPEG recompression levels used in real-world applications. The presented results have already demonstrated the effectiveness of the proposed method under representative compression conditions.

The experiments were conducted using the PyTorch framework and run on a single NVIDIA RTX 4070 GPU. The computational cost of the proposed method mainly comes from JPEG-aware preprocessing, saliency map computation, and iterative DCT-domain perturbation. The saliency map is computed using partial gradients, where only labels with higher confidence than the target label are considered, significantly reducing backpropagation overhead. Moreover, in each iteration, only a limited number of DCT coefficients are modified according to the saliency-guided strategy. Variations in JPEG quality factors do not introduce noticeable additional computational overhead. As shown in Table 5, the average runtime per image remains stable across different backbone models and compression settings, indicating good scalability for practical deployment.

In summary, the proposed method achieves the best trade-off among the four approaches, it delivers the highest attack success rate while preserving image quality and watermark integrity. This holistic superiority makes the proposed method highly suitable for adversarial scenarios involving compressed and watermarked content, where multi-objective performance is critical.

### Conclusion

In this paper, we propose a JPEG compression-resistant adversarial attack framework in which adversarial perturbations and invisible watermark information are jointly embedded directly into the DCT domain. Our experiments demonstrate that the proposed method generates adversarial examples with superior compression resistance and better visual quality compared to traditional adversarial attack methods. The proposed method can be applied by users on social networking platforms to process uploaded images, achieving robust copyright protection while simultaneously preventing unauthorized models from

scraping and classifying the images. However, the current framework is designed for standard JPEG compression and may require adaptation to handle other image coding formats or more aggressive compression settings. Future work will focus on extending the proposed framework to other compression standards and multimedia formats, improving computational efficiency, and investigating its applicability to broader vision tasks and black-box attack scenarios.

## Declarations

Ethics and data usage approval was obtained for this research. All authors consented to publication.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

## Funding

This work was supported in part by the funds of MOE (Ministry of Education in China) Project of Humanities and Social Sciences (grant number 25YJA880044), Project of Industry-university-institute Cooperation in Colleges and Universities in Fujian Province (grant number 2024h6007, 2024h6030), the Project of Software Technology Innovation Research and Industrialization in Fujian Province (grant number 2024.488).

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